Interpolare Lagrange

1. Forma clasica

$$f(x) = L_m f + R_m f$$

$$L_m f(x) = \sum_{k=0}^{m} l_k(x) f(x_k)$$

$$l_k(x) = \prod_{j=0\,;\,j\neq k}^{m} \frac{x - x_j}{x_k - x_j}$$

$$R_m f(x) = \frac{u(x)}{(m+1)!} f^{(m+1)}(\xi)$$

$$u(x) = \prod_{j=0}^{m} (x - x_j)$$

$$L_m f(x) = \sum_{k=0}^{m} \prod_{j=0;\,j\neq k}^{m} \frac{x - x_j}{x_k - x_j} f(x_k)$$

```
function fi = Lagrange(x,y,xi)
    if nargin ~=3
        error('Illegal number of arguments');
    end
    [mu,nu] = size(xi);
    fi = zeros(mu,nu);
    np1 = length(y);
    for i = 1:np1
        z = ones(mu,nu);
        for j=[1:i-1,i+1:np1]
            z = z.*(xi-x(j))/(x(i) - x(j));
        end
        fi = fi+z*y(i);
    end
end
```

3. Interpolare Lagrange baricentrica

$$L_m f(x) = \sum_{k=0}^{m} l_k(x) f(x_k)$$

$$l_k(x) = \frac{u(x)}{(x - x_k)u'(x_k)}$$

$$u'(x_k) = \prod_{j=0\,;\,j\neq k}^{m} x_k - x_j$$

$$\frac{1}{u'(x_k)} = w_k - \text{ponderi barincentrie} \Rightarrow w_k = \frac{1}{\prod_{j=0;j\neq m}^{m} x_k - x_j}$$

$$L_m f(x) = \sum_{k=0}^{m} \frac{u(x)w_k}{x - x_k} f(x_k) = u(x) \sum_{k=0}^{m} \frac{w_k}{x - x_k} f(x_k) = \frac{u(x) \sum_{k=0}^{m} \frac{w_k}{x - x_k} f(x_k)}{u(x) \sum_{k=0}^{m} \frac{w_k}{x - x_k}}$$

Tema facultativa: Interporale Lagrange fara for-uri, ponderi baricentrice fara foruri

```
function ff=barycentricInterpolation(x,y,xx,c)
    %BARYCENTRICINTERPOLATION - barycentric Lagrange interpolation
    %call ff=barycentricInterpolation(x,y,xx,c)
    %x - nodes
    %y - function values
    %xx - interpolation points
    %c - barycentric weights
    %ff - values of interpolation polynomial
    n=length(x)-1;
    numer = zeros(size(xx));
    denom = zeros(size(xx));
    exact = zeros(size(xx));
    for j=1:n+1
    xdiff = xx-x(j);
    temp = c(j)./xdiff;
    numer = numer+temp*y(j);
    denom = denom+temp;
    exact(xdiff==0) = j;
    end
    ff = numer ./ denom;
    jj = find(exact);
    ff(jj) = y(exact(jj));
end

function c = barycentricweigths( x )
    %BARYCENTRICWEIGHTS - compute barycentric weights(coefficient)
    %call c = barycentricweigths( x )
    %x - nodes
    %c - weights
```

```
    n=length(x)-1;
    c=ones(1,n+1);
    for j=1:n+1
    c(j)=prod(x(j)-x([1:j-1,j+1:n+1]));
    end
    c=1./c;
end

function ff=ChebLagrange2(y,xx,a,b)
    %CHEBLAGRANGE - Lagrange interpolation for Chebyshev #2 points- barycentric
    %call ff=ChebLagrange(y,xx,a,b)
    %y - function values;
    %xx - evaluation points
    %a,b - interval
    %ff - values of Lagrange interpolation polynomial
    n = length(y)-1;
    if nargin==2
        a=-1; b=1;
    end
    c = [1/2; ones(n-1,1); 1/2].*(-1).^((0:n)');
    x = sort(cos((0:n)'*pi/n))*(b-a)/2+(a+b)/2;
    ff=barycentricInterpolation(x,y,xx,c);
end

function ff=ChebLagrange1(y,xx,a,b)
%CHEBLAGRANGE1 - Lagrange interpolation for Chebyshev #1 points- barycentric
%call ff=ChebLagrangek1(y,xx,a,b)
%y - function values;
%xx - evaluation points
%a,b - interval
%ff - values of Lagrange interpolation polynomial

n = length(y)-1;
if nargin==2
    a=-1; b=1;
end
c = sin((2*(0:n)'+1)*pi/(2*n+2)).*(-1).^((0:n)');
x = sort(cos((2*(0:n)'+1)*pi/(2*n+2))*(b-a)/2+(a+b)/2);
ff=barycentricInterpolation(x,y,xx,c);
end
```

4. Probleme

Problema 1

Interpolare Lagrange

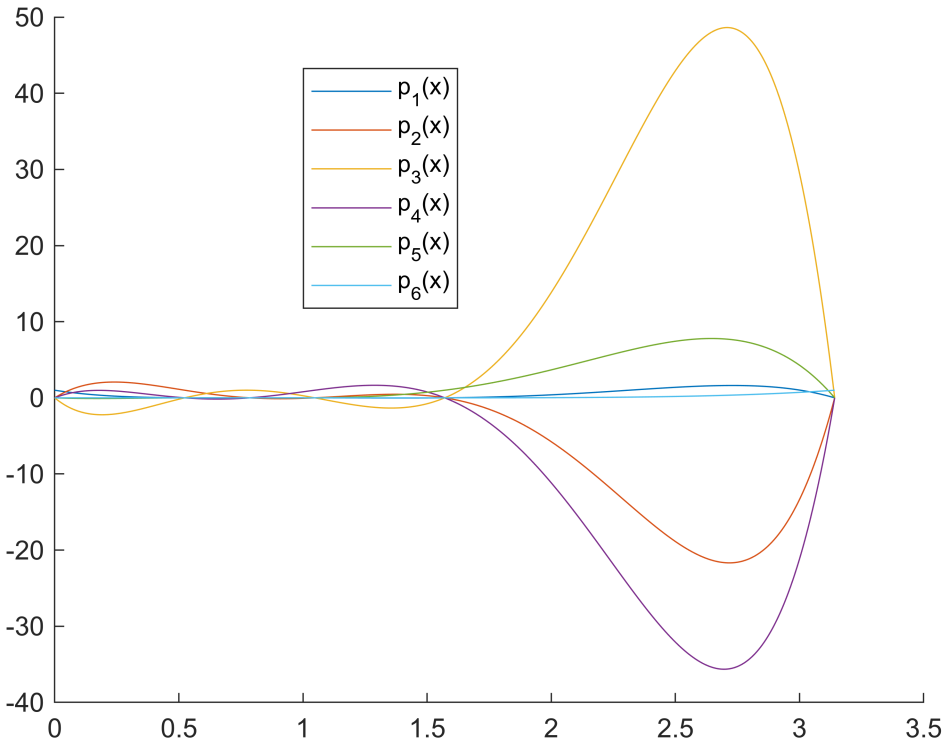
Problema 2

```
nod = [0, pi/6, pi/4, pi/3, pi/2, pi];
syms x;
np1 = numel(nod);

hold on;
labels = strings(1, np1);

for i = 1:np1
    z = 1;
    for j = setdiff(1:np1, i)
        z = z * (x - nod(j)) / (nod(i) - nod(j));
    end
    vals = linspace(0, pi, 1000);
    plot(vals, double(subs(z, x, vals)));
    labels(i) = sprintf('p_%d(x)', i);
end

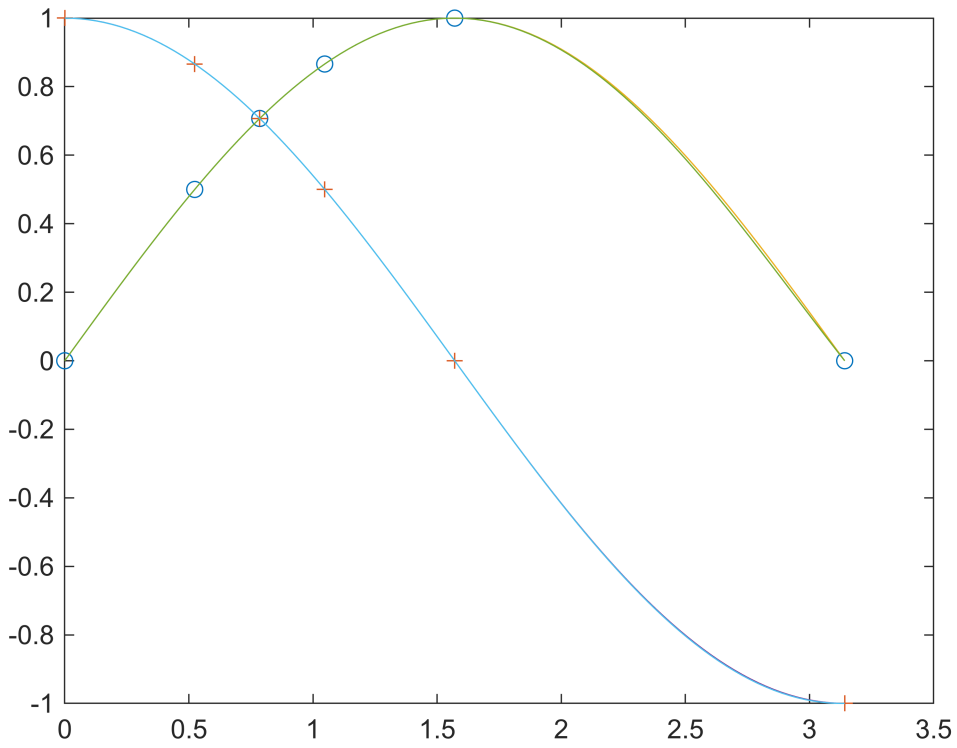
legend(labels, 'Location', 'best');
hold off;
```



Problema 3 si 4

```
x = [0,pi/6,pi/4,pi/3,pi/2,pi];
y_sin = [0,1/2,sqrt(2)/2,sqrt(3)/2,1,0];
y_cos = [1,sqrt(3)/2,sqrt(2)/2,1/2,0,-1];
t=linspace(0,pi,100);
P_sin = Lagrange(x,y_sin,t);
```

```
P_cos = Lagrange(x,y_cos,t);
plot(x, y_sin, 'o', x, y_cos, '+', t, sin(t),t,cos(t),t,P_sin,t,P_cos);
```



Problema 5

Implementare cu puncte Cebisev

5. Probleme practice

Problema 1

```
x = 1900:10:2010
```

```
x = 1×12
        1900        1910        1920        1930        1940        1950 ···
```

```
y=[75995,91972,105711,123203,131669,150697,179323,203212,226505,249633,281422,308786
]
```

```
y = 1×12
       75995       91972      105711      123203      131669      150697 ···
```

```
format long
P1 = Lagrange(x,y,1975)
```

```
P1 =
     2.143643887577056e+05
```

```
P2 = Lagrange(x,y,2018)
```

```
P2 =
    -6.643068132291096e+05
```

Problema 2

```
f = @(x) exp(x.^2-1);
x=[1,1.1,1.2,1.3,1.4];
y=f(x);
P = Lagrange(x,y,1.25)
```

```
P =
   1.754957328862147
```

Problema 3

```
y=10:0.1:11;
x=y.*y;
P = Lagrange(x,y,115)
```

```
P =
  10.723805294763606
```

```
abs(P-sqrt(115))
```

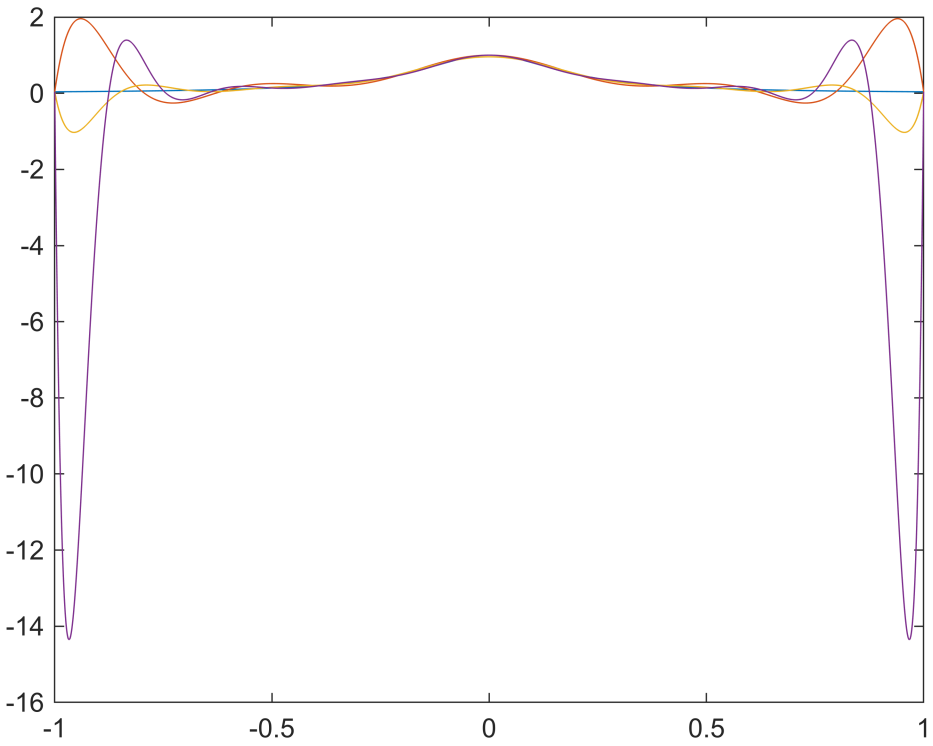
```
ans =
     1.776356839400250e-15
```

Problema 4

Contraexemplul lui Runge

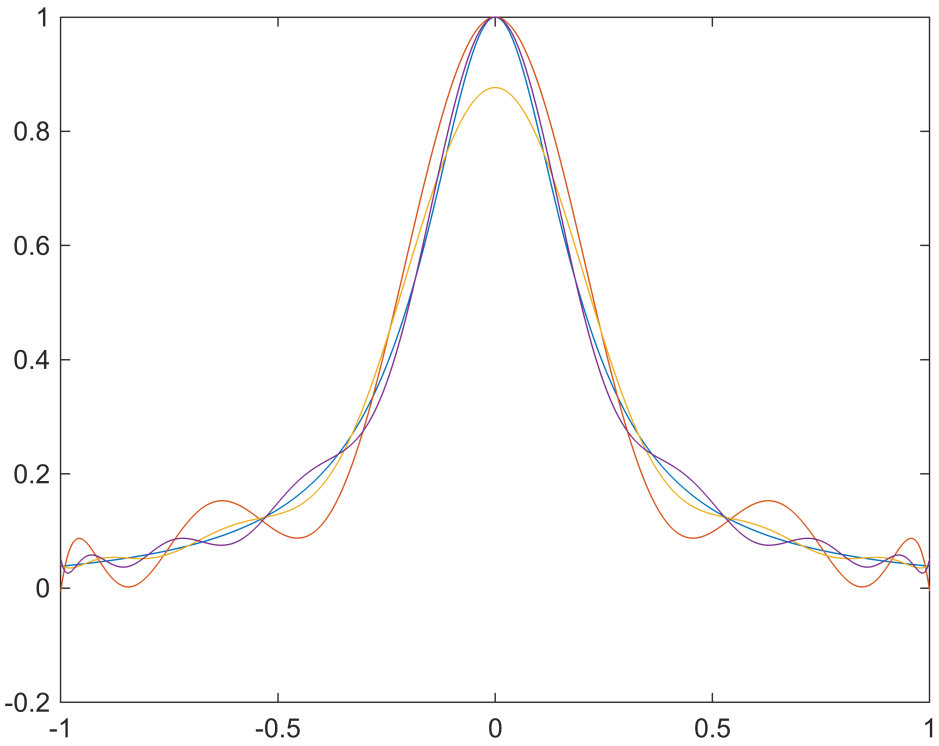
Noduri echidistante

```
f = @(x) 1./(1+25*x.^2);
tg = linspace(-1,1,3000)';
yg = f(tg);
L = zeros(length(yg),3);
k=0;
for m = [10,13,16]
    xn = linspace(-1,1,m+1);
    yn = f(xn);
    k=k+1;
    c = barycentricweigths(xn);
    L(:,k) = barycentricInterpolation(xn,yn,tg,c);
end
plot(tg,[yg,L])
```



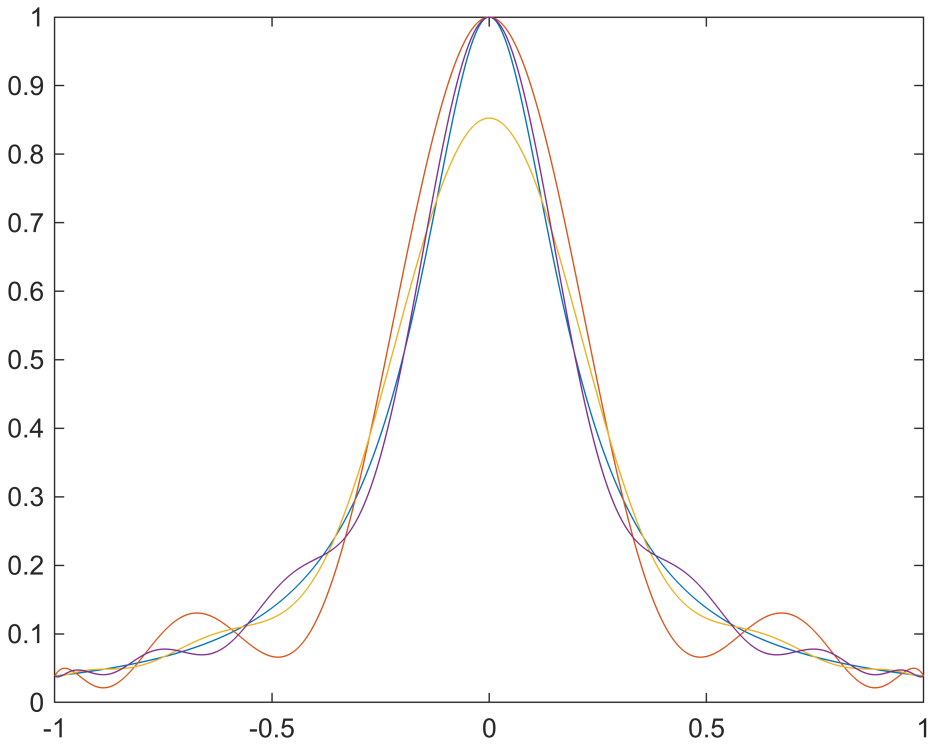
Cebisev speta #1

```
f = @(x) 1./(1+25*x.^2);
tg = linspace(-1,1,3000)';
yg = f(tg);
L = zeros(length(yg),3);
k=0;
for m = [10,13,16]
    xn = sort(cos((2*(0:m)'+1)*pi/(2*m+2)));
    yn = f(xn);
    k=k+1;
    L(:,k) = ChebLagrange1(yn,tg);
end
plot(tg,[yg,L])
```



Cebisev speta #2

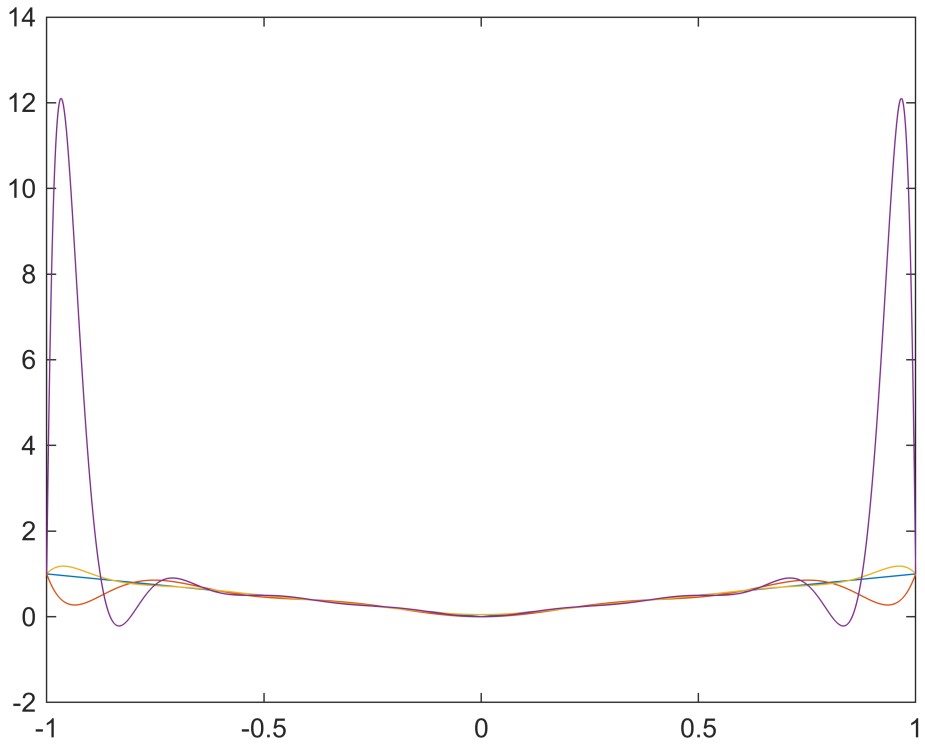
```
f = @(x) 1./(1+25*x.^2);
tg = linspace(-1,1,3000)';
yg = f(tg);
L = zeros(length(yg),3);
k=0;
for m = [10,13,16]
    xn = sort(cos((0:m)'*pi/m));
    yn = f(xn);
    k=k+1;
    L(:,k) = ChebLagrange2(yn,tg);
end
plot(tg,[yg,L])
```



Contraexemplul lui Bernstein

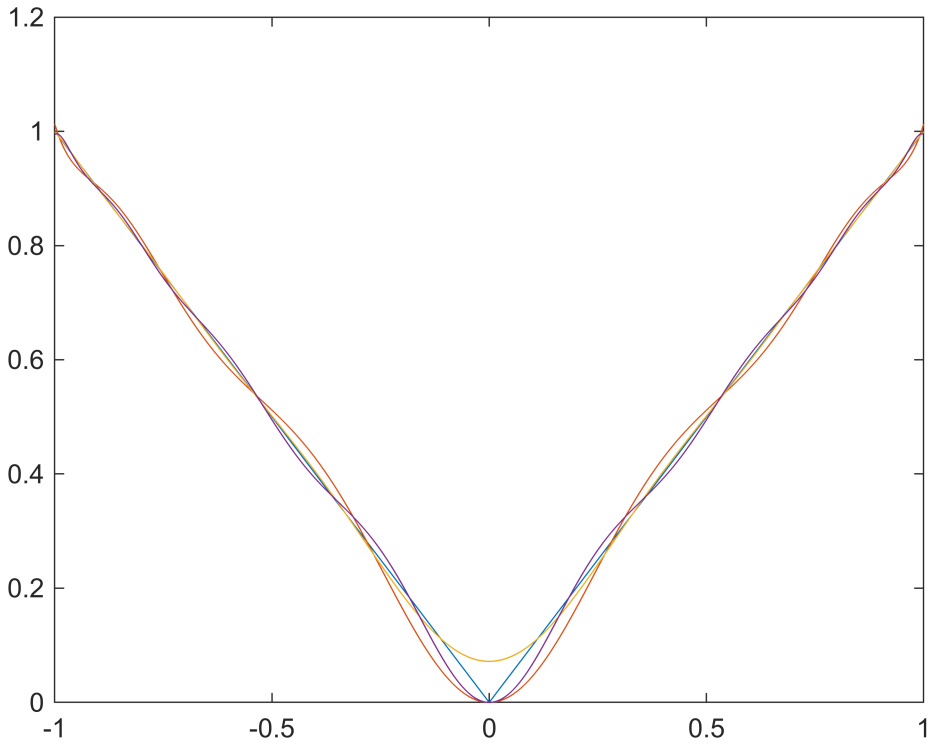
Noduri echidistante

```
f = @(x) abs(x);
tg = linspace(-1,1,3000)';
yg = f(tg);
L = zeros(length(yg),3);
k=0;
for m = [10,13,16]
    xn = linspace(-1,1,m+1);
    yn = f(xn);
    k=k+1;
    c = barycentricweigths(xn);
    L(:,k) = barycentricInterpolation(xn,yn,tg,c);
end
plot(tg,[yg,L])
```



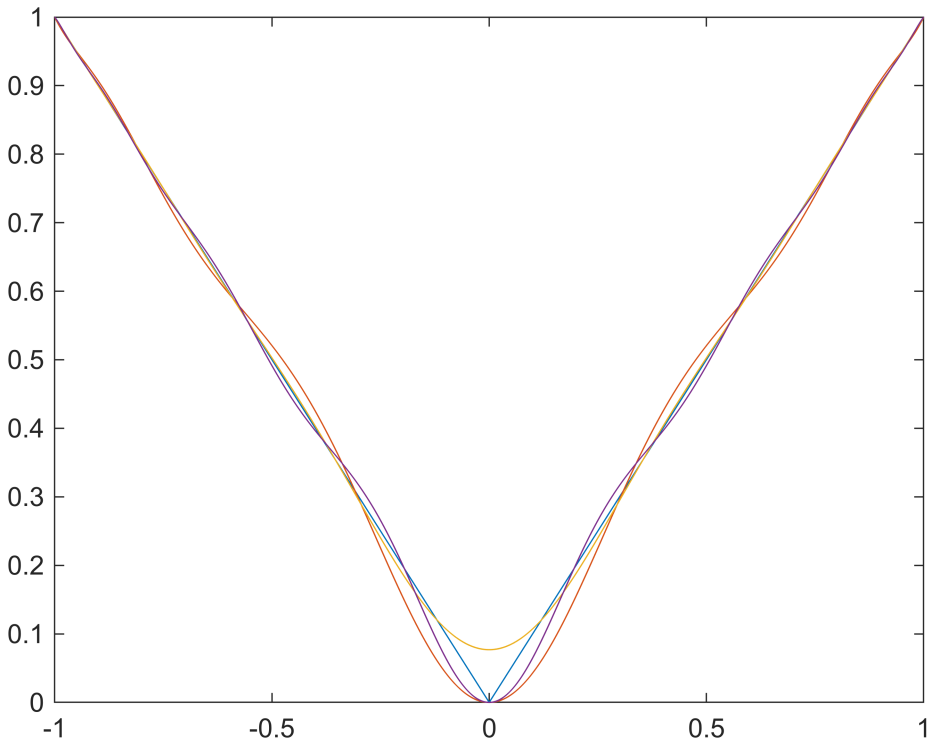
Cebisev speta #1

```
f = @(x) abs(x);
tg = linspace(-1,1,3000)';
yg = f(tg);
L = zeros(length(yg),3);
k=0;
for m = [10,13,16]
    xn = sort(cos((2*(0:m)'+1)*pi/(2*m+2)));
    yn = f(xn);
    k=k+1;
    L(:,k) = ChebLagrange1(yn,tg);
end
plot(tg,[yg,L])
```



Cebisev speta #2

```
f = @(x) abs(x);
tg = linspace(-1,1,3000)';
yg = f(tg);
L = zeros(length(yg),3);
k=0;
for m = [10,13,16]
    xn = sort(cos((0:m)'*pi/m));
    yn = f(xn);
    k=k+1;
    L(:,k) = ChebLagrange2(yn,tg);
end
plot(tg,[yg,L])
```



Problema 5

```
x = [0,pi/6,pi/4,pi/3,pi/2];
y_sin = [0,1/2,sqrt(2)/2,sqrt(3)/2,1];
y_cos = [1,sqrt(3)/2,sqrt(2)/2,1/2,0];
P_sin=Lagrange(x,y_sin,pi/36)
```

```
P_sin =
   0.086913410889062
```

```
P_cos=Lagrange(x,y_cos,pi/36)
```

```
P_cos =
   0.996386195987906
```

```
abs(cos(pi/36) - P_cos)
```

```
ans =
     1.914978961601310e-04
```

```
abs(sin(pi/36) - P_sin)
```

```
ans =
     2.423318585964285e-04
```